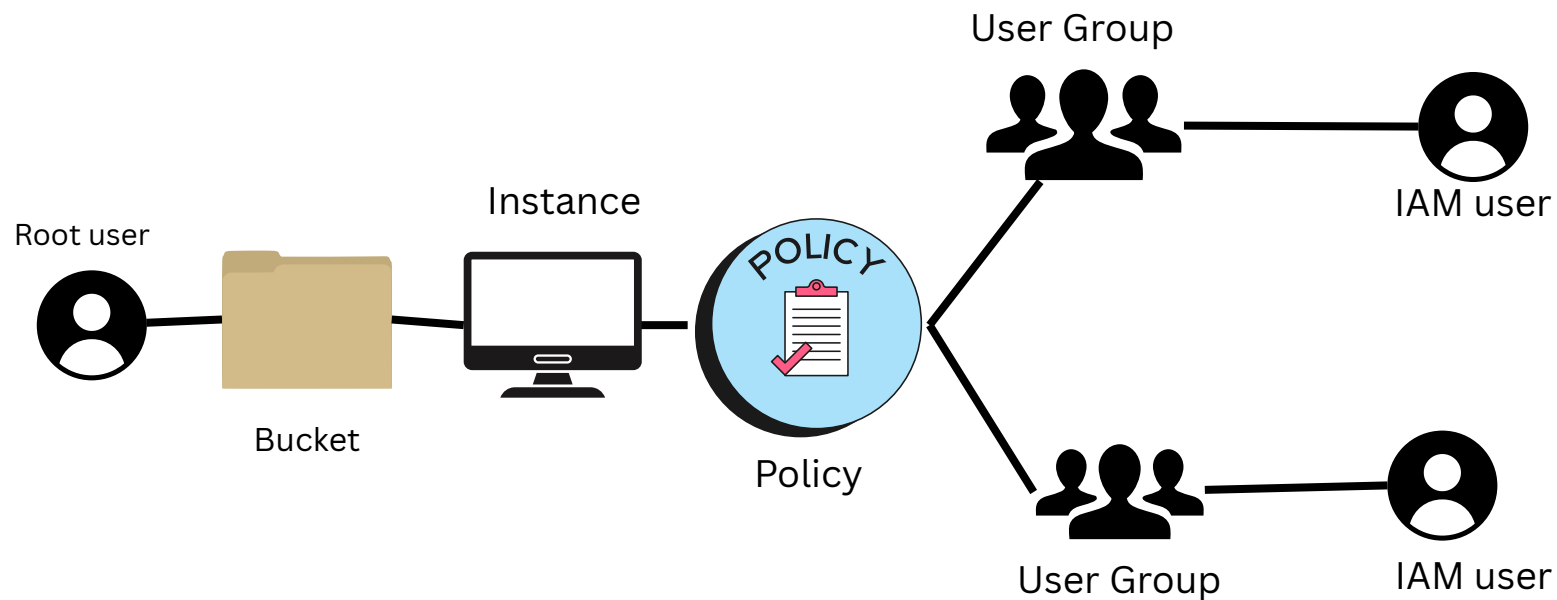


AWS IAM Cloud Security Project

1. Project Overview

I completed this project on cloud security controls in Amazon Web Services (AWS), focusing on Identity and Access Management (IAM). The goal was to create a least-privilege policy, attach it to a users, and verify that the policy correctly restricts actions on S3 bucket.

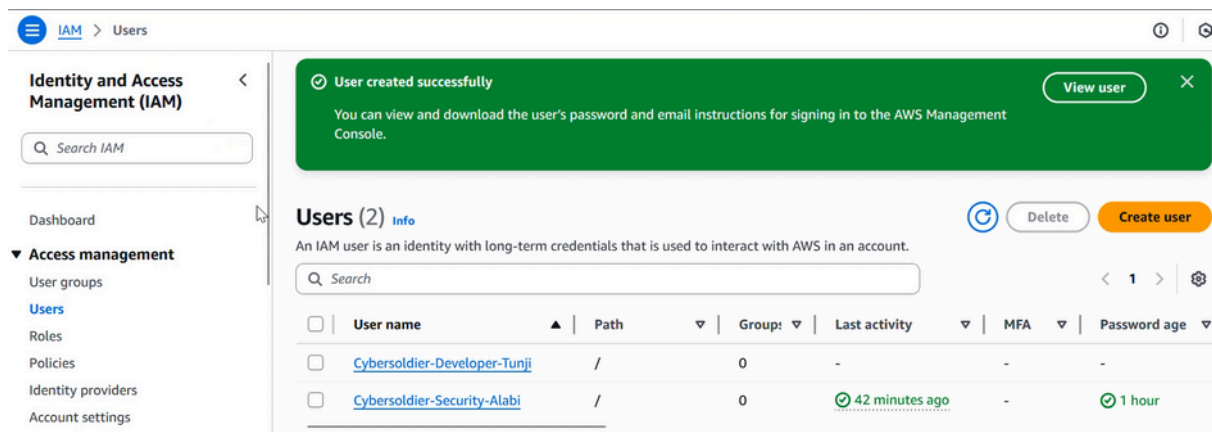


2. Tools & Concepts

- AWS IAM – users, groups, policies.
- Amazon EC2 – instance and tags.
- JSON policy syntax – Effect, Action, Resource
- Principle of least privilege and policy testing

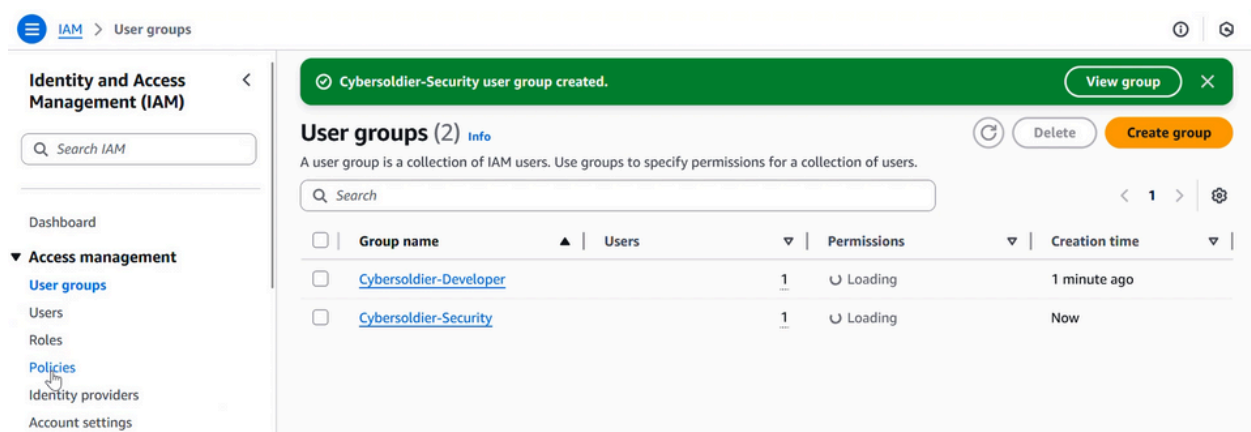
3. IAM Users

Created IAM users cybersoldier-secutiry-alabi & cybersoldier developer-tunji.



4. IAM Users & Groups

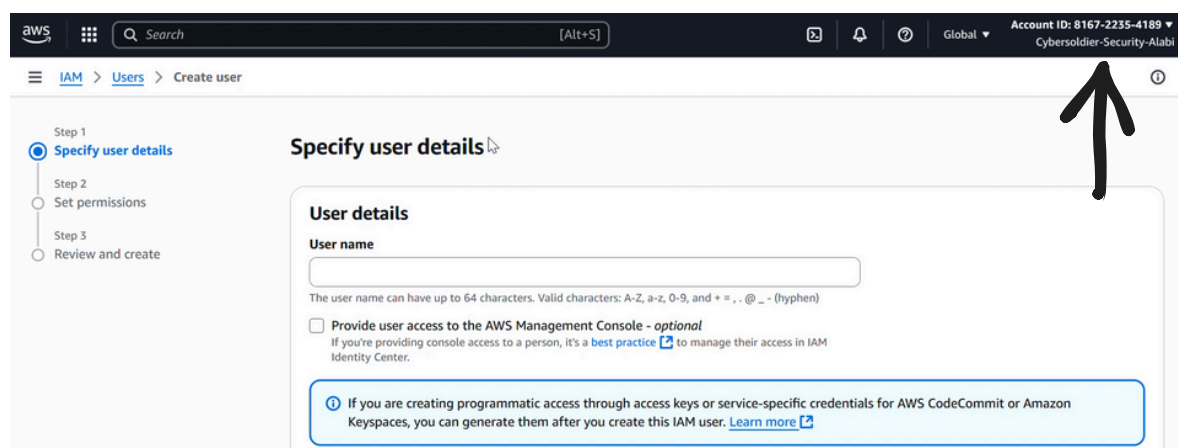
1. Created an IAM user group called cybersoldier-security & cybersoldier-developer.
2. Attached alabi to the security group and also added tunji to the developer group.



5. Logging in as an IAM User

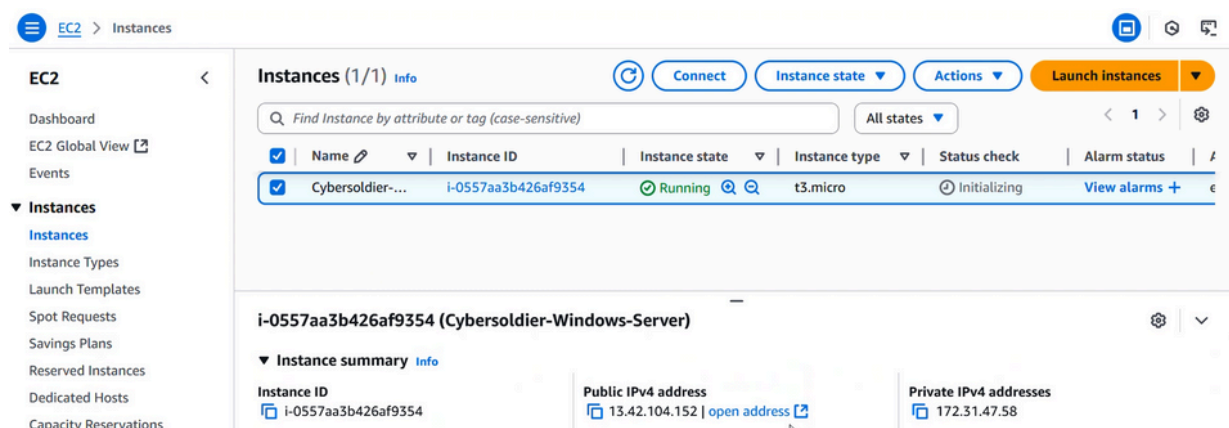
IAM users can sign in through:

- AWS Management Console (using the new alias URL)
- AWS CLI via programmatic keys



6. Instance

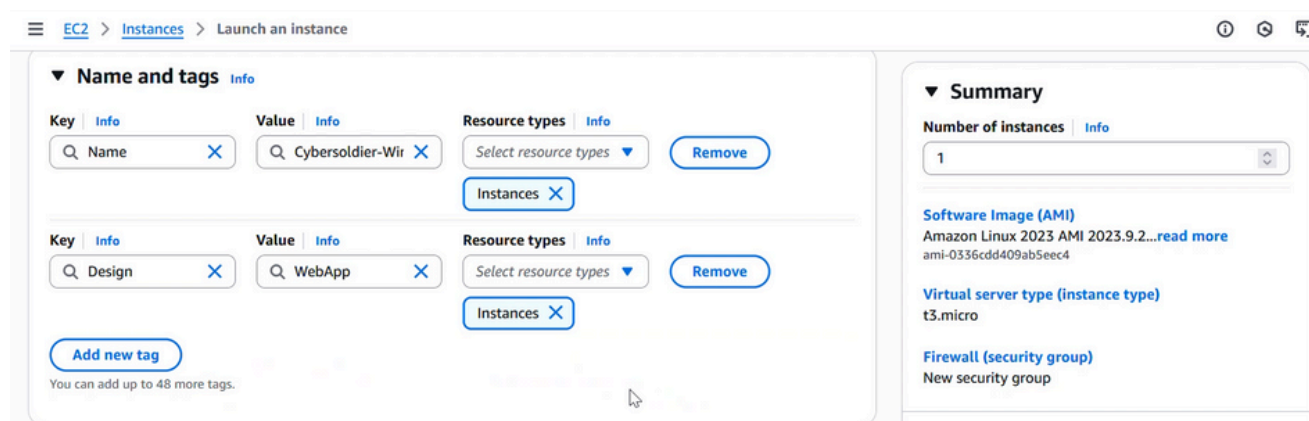
Created an instance with the name Cybersoldier-Windows-server using the EC2 service



3. Tagging Strategy

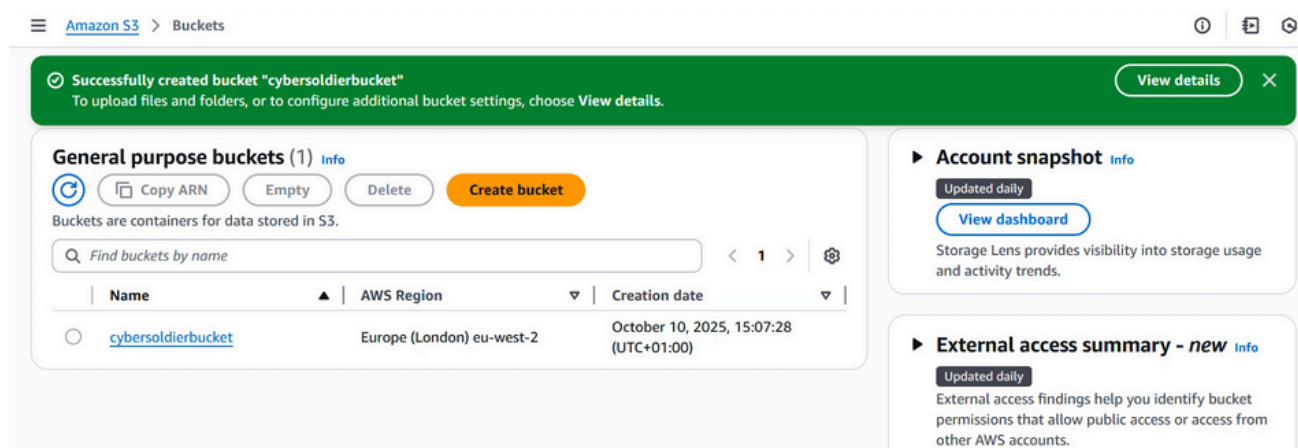
I applied a descriptive tag to the EC2 instance:

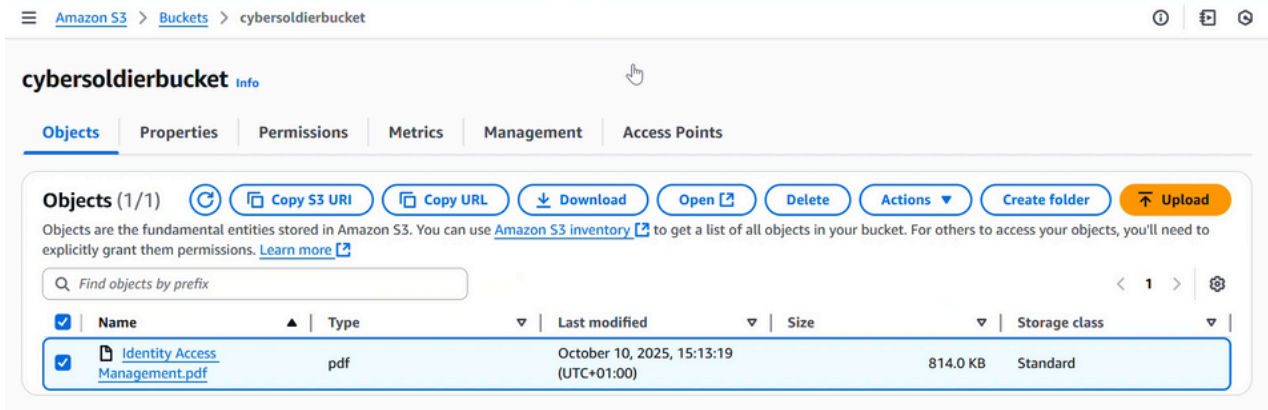
Instance |Name and tags| Key Design | Tag Value Webapp.



6. Bucket

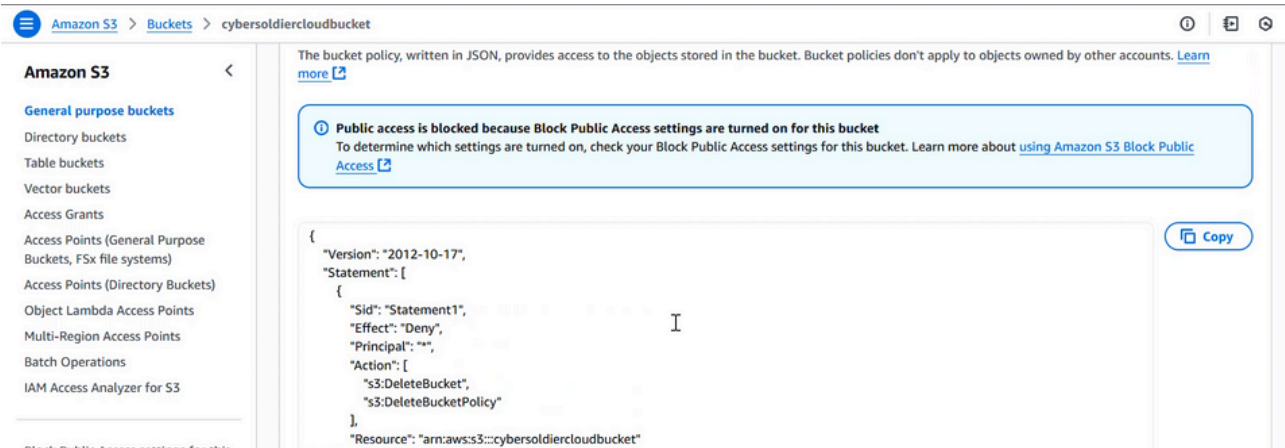
Created a bucket with the name cybersoldierbucket using the S3 service and also upload an object into the bucket





4. Creating the S3 Bucket Policy

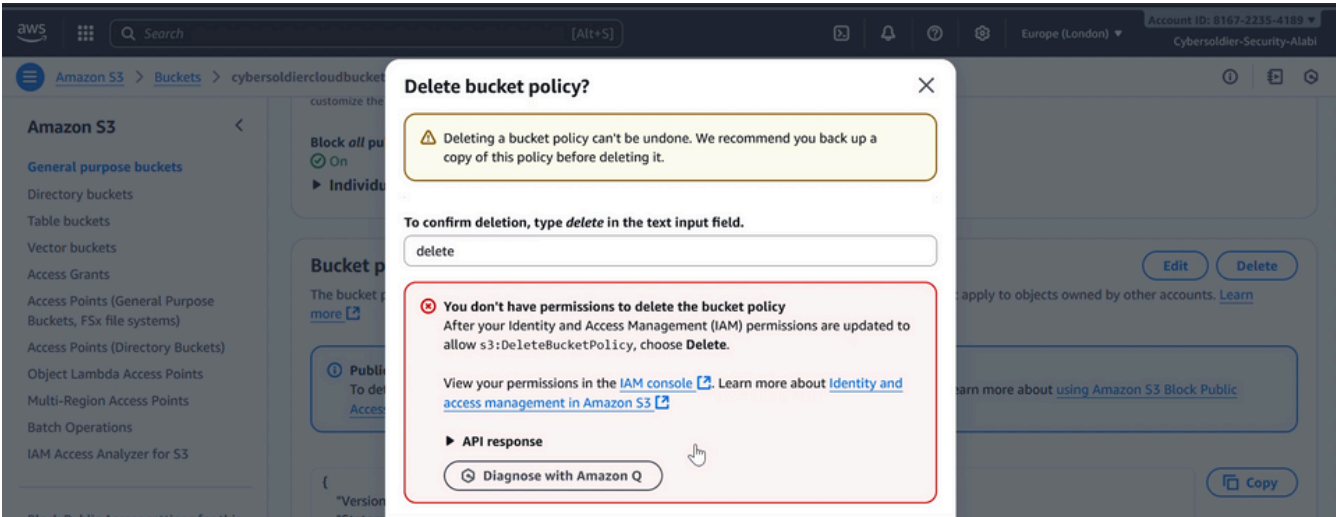
I authored the following JSON policy to deny IAM users access to delete the bucket or delete bucket policy.



8. Testing the Policy

Test Action | Expected Result | Actual Result

Delete bucket/Delete bucket policy | Denied | No permission to delete bucket or bucket policy



In conclusion, this project have successfully demonstrated the implementation of cloud security measures on Amazon Web Services(AWS). By leveraging AWS security features which effectively protect cloud infrastructure and data from potential threats. This project provide valuable insight for organization seeking to enhance their cloud security posture. As cloud computing continue to evolve, this project might serve as a foundation for future security initiatives, ensuring Confidentiality, Integrity and Availablity of sensitive data in cloud.